

Q1: 25 Feb (Shift 1) - Numerical

Using the provided information in the following paper chromatogram:

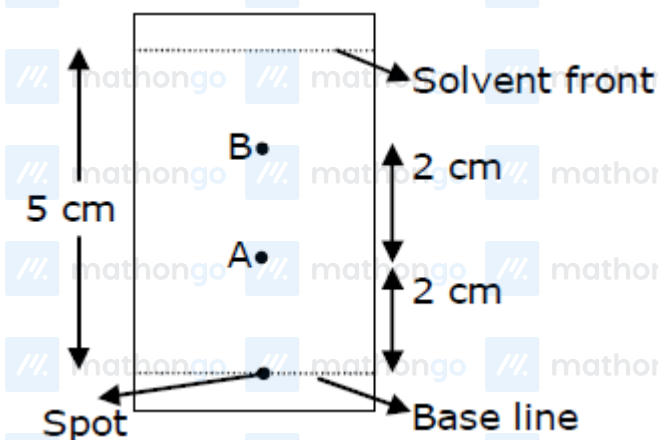


Fig: Paper chromatography for compounds A and B

The calculated R_f value of A $\times 10^{-1}$

Q2: 26 Feb (Shift 1) - Single Correct

Which of the following is 'a' FALSE statement?

- (1) Carius tube used in the estimation of sulphur in an organic compound
- (2) Kjeldahl's method is used for the estimation of nitrogen in an organic compound
- (3) Phosphoric acid produced on oxidation of phosphorus present in an organic compound is precipitated as $Mg_2P_2O_7$ by adding magnesia mixture
- (4) Carius method is used for the estimation of nitrogen in an organic compound

Q3: 26 Feb (Shift 1) - Single Correct

Given below are two statements :

Statement I : A mixture of chloroform and aniline can be separated by simple distillation

Statement II : When separating aniline from a mixture of aniline and water by steam distillation aniline boils below its boiling point

In the light of the above statements, choose the most appropriate answer from the options given below

Questions with Answer Keys

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(1) Statement I is true, statement II is false

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is false, Statement II is true

Answer Key

Q1 (4)

Q2 (4)

Q3 (2)

Hints and Solutions

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Q1 (4)

$$R_f = \frac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$$

On chromatogram distance travelled by compound is $\rightarrow 2$ cm

Distance travelled by solvent = 5 cm

$$\text{So } R_f = \frac{2}{5} = 0.4$$

Q2 (4)

Fact

Q3 (2)

A suitable method for separating a mixture of aniline and chloro form would be steam distillation. Steam distillation is the process used to separate aromatic compound from a mixture because of their temperature sensitivity. Therefore, steam distillation is an ideal method for their separation.